

Analysis Ethical Considerations Report

Assignment 15: Loan Default Prediction

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GitHub Link:

https://github.com/AyleenSantander/M.11_Intro_AI/blob/main/M11_A_15_Loan_Default_Prediction.ipynb

1. Model Summary

The model predicts whether a loan applicant will default, using 700 labelled records and 9 features. About 26% of applicants defaulted. The Random Forest scored 0.83 ROC AUC on the test set with 81% accuracy, but recall on defaults was only 0.49, it misses about half the people who actually default. Logistic Regression scored higher in cross-validation (0.842 versus 0.797).

2. Fairness

This dataset has no gender column, so I couldn't check gender fairness. I checked age and education instead, since both are protected characteristics in lending.

I split applicants into age groups and compared how often the model flagged each group, and how often it got them wrong. I included each group's real default rate too, younger applicants really do default more here, so flagging them more isn't automatically unfair. The question is whether the model flags them more than the real rate justifies.

Model performance by age group

Age group	n	Actual default rate	Flagged as risky	TPR	FPR
Under 30	44	0.364	0.295	0.562	0.143
30 – 39	63	0.206	0.159	0.538	0.060
40 – 49	27	0.185	0.074	0.400	0.000
50 +	6	0.500	0.000	0.000	0.000

TPR = share of real defaulters correctly caught. FPR = share of good applicants wrongly flagged.

The results showed a real gap. Applicants under 30 had a false positive rate of 0.143, compared with 0.060 for ages 30–39 and 0.000 for 40–49. False positives matter most in lending, because a false positive means a good applicant gets denied a loan. Education level 2 showed the same problem: a 0.147 false positive rate and precision of only 0.44, versus 0.02 and 0.90 for level 1.

The 50+ group flagged nobody at all and missed all 3 of its actual defaults. But that group has only 6 people in the test set, so this number doesn't mean much, it's also what dragged the selection rate ratio down to 0.000, which looks alarming but is really just a sample size problem.

One important point: dropping age from the model wouldn't fix age bias, because employ and address are tied to age and act as stand-ins for it. So, I measured the results instead of assuming removal solved it.

3. Privacy

There are no names or ID numbers in this data, so it's already anonymous on the surface. But age, education, income and job tenure combined could still identify someone in a small group.

In a real deployment I would: only collect fields the model actually uses, encrypt the file, limit who can open it, delete it after a set period, and protect the predictions too, a risk score attached to a person is personal data just like the inputs.

Under Canadian privacy law (PIPEDA), using loan data to build risk profiles is a new purpose, so it needs its own consent. And if someone withdraws consent, deleting their row isn't enough, the trained model still carries their influence, so it must be retrained.

4. Transparency

I used three methods because they answer different questions:

- **Feature importance** = which features the model relies on
- **Permutation importance** = how much performance drops when a feature is scrambled (more reliable)
- **SHAP** = shows how each feature pushes a prediction, for the whole model and for one applicant at a time

All three agreed on the top factors: debt-to-income, years employed, and credit card debt. Permutation importance also showed that income, other debt, and most education levels contribute nothing reliable, their error bars cross zero.

Logistic regression coefficients gave the direction. Each standard deviation more credit card debt multiplies the odds of default by 3.22; each standard deviation more employment tenure multiplies them by 0.18. Debt-to-income sits at 1.55.

The SHAP waterfall plot is the important one: it explains why this specific person got flagged. Lenders legally must give a reason when they deny someone, so that matters. The dependence plot also showed that risk stays flat until debt-to-income crosses roughly 10–15, then climbs sharply, useful for setting a screening cutoff.

Two honest limits: SHAP explains what the model thinks, not what's true. And these are correlations, not causes, staying in a job longer wouldn't actually make someone less likely to default.

Human oversight

The model is a helper, not the decision-maker. With recall at 0.49, it isn't accurate enough to decide anything alone. A person should review every denial, using the SHAP explanation as supporting evidence.

5. Improvements to Address the Identified Bias

1. **Get more data.** 700 rows with 183 defaults are too small. Some age groups have under 30 people in the test set, so the fairness numbers jump around, and you can't tell a real problem from noise. Cross-validated predictions help meanwhile.
2. **Stop using 0.5 as the cutoff.** Denying a good customer and approving a bad one cost the bank different amounts, so pick a cutoff that matches those costs.
3. **Check whether other columns act like age.** Employment and address both rise with age, so deleting the age column wouldn't stop the model from guessing it.
4. **Fix the class imbalance.** `class_weight='balanced'` helps; SMOTE is the next step if defaults are still being missed. SMOTE goes inside the pipeline so it only touches training data.
5. **Different cutoffs per group, carefully.** This equalizes error rates but means treating people differently based on age, which is legally risky in lending.
6. **Prefer logistic regression.** It scored higher than Random Forest in cross-validation and its coefficients can be read directly, which makes explaining a denial much easier.
7. **Keep checking after launch.** Bias appears over time as the applicant pool changes. Re-run the audit every few months on real decisions.
8. **Have a person review every denial.** The most reliable fix with data is this small.

Final note: with only 700 rows, none of these fixes could be proven to work. Saying that honestly is more useful than claiming the bias was solved.